

GNUBG Unity Installer (UPM)

Unity Package Manager installer that downloads GNUBG binaries (Windows/macOS/Linux) from GitHub Releases and unpacks them into Assets/StreamingAssets/gnubg. You don't *have* to, but it's the cleanest setup:

This repository contains a Unity Package Manager (UPM) package that installs GNU Backgammon (GNUBG) runtime binaries into a Unity project.

On install (or via menu), the package downloads prebuilt GNUBG bundles for **Windows / macOS / Linux** from a separate GitHub Releases repository, then unzips them into:

```
Assets/StreamingAssets/gnubg/  
windows/  
macos/  
linux/
```

This layout is designed to support a Unity bridge that launches `gnubg-cli` from `StreamingAssets`.

What this package does

- Fetches the **latest GitHub Release** metadata from a configured builds repo
- Downloads these assets:
 - `gnubg-windows.zip`
 - `gnubg-macos.zip`
 - `gnubg-linux.zip`
- Extracts each zip to `Assets/StreamingAssets/gnubg/<platform>/`
- Normalises the executable name to:
 - Windows: `gnubg-cli.exe`
 - macOS/Linux: `gnubg-cli`
- Ensures executable permissions on macOS/Linux (Editor-side)

Requirements

- Unity 2021.3+ recommended
- Internet access during installation/update (Editor)
- A builds repository that publishes the GNUBG zip assets via GitHub Releases

Install (Unity Package Manager)

Unity Package Manager → Add package from git URL...

Use:

```
[https://github.com/] (https://github.com/)<OWNER>/<REPO>.git?path=/Packages/com.stonesanddice
```

Replace <OWNER> and <REPO> with this repository.

Configure the builds repository

The installer needs to know where to download GNUBG binaries from.

Edit:

```
Packages/com.stonesanddice.gnubg.installer/Editor/GnubgInstaller.cs
```

Set:

- `GitHubOwner`
- `GitHubRepo`

The builds repo must publish the release assets:

- `gnubg-windows.zip`
 - `gnubg-macos.zip`
 - `gnubg-linux.zip`
-

Install / Update binaries

Menu:

Tools → GNUBG → Install or Update Binaries

This will download and unpack all platforms into `Assets/StreamingAssets/gnubg/`.

Artefact layout expected inside each zip

Each zip should contain a single top-level folder named by platform, or just the contents directly. The installer can flatten one extra platform folder level if needed.

Recommended zip contents:

```
windows/  
gnubg-cli.exe  
*.dll
```

```
share/gnubg/...  
lib/python3.12/...    (if embedding python)  
macos/  
gnubg-cli  
share/gnubg/...  
linux/  
gnubg-cli  
share/gnubg/...
```

Notes for embedded Python (Windows)

If your GNUBG build embeds Python, ensure the Windows zip includes:

- `lib/python3.12/` (including `encodings/`)
- `lib/python312.zip` (if required by your build)

If your runtime sets `PYTHONHOME`/`PYTHONPATH`, do so relative to the extracted folder.